Supplemental Material: A Common Recovery Transition for Classical and Quantum Information in Random Codes

OpenQuantum
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TASKS, NORMALIZATION, AND RANDOMNESS

Let $d = 2^n$, $2 \le M \le d$, and let $V_0 : \mathbb{C}^M \to \mathbb{C}^d$ be a fixed isometry. Choose U from an exact unitary 2-design and set $V = UV_0$. The logical basis $\{|j\rangle\}_{j=1}^M$ is fixed before drawing U. The ensemble may depend on n and need not be Haar. The Pauli channel

$$\mathcal{N}_p(\rho) = \sum_{P \in \{I,X,Y,Z\}^{\otimes n}} p_P P \rho P^\dagger \tag{S1}$$

is fixed independently of U; spatially correlated probabilities are allowed. For zero-probability errors define the surprisal as $+\infty$, and set

$$s(P) = -\log_2 p_P, \qquad \mathcal{F}(t) = \mathbb{P}_{P\sim p}[s(P) < t], \qquad k = \log_2 M. \tag{S2}$$

Each decoder knows the realized encoder V and the fixed Pauli channel. Optimization is performed separately for each encoder before ensemble averaging.

For $\rho_j = \mathcal{N}_p(V|j\rangle\langle j|V^\dagger)$, the classical task is

$$P_{\mathrm{c}}^*(V) = \max_{\{E_j\}} \frac{1}{M} \sum_j \mathrm{tr}(E_j \rho_j), \qquad E_j \ge 0, \quad \sum_j E_j = I_d. \tag{S3}$$

The quantum task is

$$F_{\mathrm{q}}^*(V) = \max_{\mathcal{D}} \langle \Phi_M | [\mathrm{id} \otimes (\mathcal{D} \circ \mathcal{N}_p \circ \mathcal{V})](\Phi_M) | \Phi_M \rangle, \quad |\Phi_M\rangle = M^{-1/2} \sum_j |j\rangle|j\rangle, \quad \mathcal{V}(\rho) = V\rho V^\dagger. \tag{S4}$$

The decoder $\mathcal{D} : \mathbb{C}^d \to \mathbb{C}^M$ is any completely positive trace-preserving map. Fidelity is the overlap, not its square root, and the two tasks can have different optimal decoders. The model uses ideal encoding and places no computational constraint on decoding. Its performance criteria are uniform complete-message success and entanglement fidelity, rather than worst-input or diamond-norm recovery; the encoder ensemble is prescribed rather than optimized for channel capacity.

TWO EXACT SECOND-MOMENT IDENTITIES

For a fixed logical column $|\psi_j\rangle = V|j\rangle$,

$$\mathbb{E}|\psi_j\rangle\langle\psi_j|^{\otimes 2} = \frac{I + F}{d(d+1)}. \tag{S5}$$

For any traceless unitary A and orthogonal logical columns $i \ne j$,

$$\mathbb{E}|\langle\psi_j|A|\psi_j\rangle|^2 = \frac{1}{d+1}, \tag{S6}$$

$$\mathbb{E}|\langle\psi_i|A|\psi_j\rangle|^2 = \frac{d}{d^2-1}. \tag{S7}$$

To see the latter coefficient, extend V_0 to a full orthonormal basis. For Haar U, the off-diagonal matrix elements of $U^\dagger A U$ have equal mean squared modulus. Their sum together with the diagonal term in one column is one. Thus the off-diagonal mean is $[1 - 1/(d+1)]/(d-1) = d/(d^2-1)$. All expressions are degree $(2,2)$ in U, U^* and therefore have the same averages for an exact unitary 2-design. For distinct Pauli labels, $P^\dagger Q$ is a traceless unitary even when its phase is $\pm i$.

For a set of errors $\mathcal{T}$, let $B_j = \sum_{P\in\mathcal{T}} p_P P|\psi_j\rangle\langle\psi_j|P^\dagger$, $m = \sum_{\mathcal{T}} p_P$, and $v = \sum_{\mathcal{T}} p_P^2$. Pauli orthogonality and Eq. (S5) give

$$\mathbb{E}\,\mathrm{tr}\,B_j^2 = \sum_{P,Q\in\mathcal{T}} p_P p_Q \mathbb{E}|\langle\psi_j|P^\dagger Q|\psi_j\rangle|^2 = v + \frac{m^2-v}{d+1} = \frac{dv+m^2}{d+1}. \tag{S8}$$

These identities retain the orthogonality correlations between code columns.

A CLASSICAL CONVERSE THAT ALSO BOUNDS QUANTUM RECOVERY

For every $c > 0$ and every POVM,

$$P_{\mathrm{succ}} \le \frac{cd}{M} + \frac{1}{M}\sum_j \mathrm{tr}(\rho_j - cI)_+. \tag{S9}$$

For a threshold $c' > 0$, write $\rho_j = A_j + B_j$, with A_j containing errors of probability $p_P > c'$ and B_j the remaining errors. Positivity gives

$$\mathrm{tr}(A_j + B_j - cI)_+ \le \mathrm{tr}\,A_j + \mathrm{tr}(B_j - cI)_+ \le \mathcal{F}(\log_2(1/c')) + \frac{\mathrm{tr}\,B_j^2}{4c}. \tag{S10}$$

The first inequality follows by maximizing $\mathrm{tr}\,Q(A_j+B_j-cI)$ over $0 \le Q \le I$; the second follows from $(\lambda-c)_+ \le \lambda^2/(4c)$ for $\lambda \ge 0$. Since $v \le c'm \le c'$ and $m \le 1$, Eq. (S8) implies $\mathbb{E}\,\mathrm{tr}\,B_j^2 \le c' + 1/d$. Put $\gamma = cd$ and $c' = c2^{-a}$ to obtain

$$\mathbb{E}P_{\mathrm{c}}^* \le \frac{\gamma}{M} + \frac{1}{4\gamma} + \frac{2^{-a}}{4} + \mathcal{F}(n - \log_2\gamma + a), \qquad \gamma > 0,\ a > 0. \tag{S11}$$

In particular, $\gamma = M2^{-a}$ yields the upper bound in the main text. Because the underlying inequality holds for every POVM, the receiver can optimize its measurement after learning the sampled code.

For any recovered logical channel with Kraus operators A_ℓ, Cauchy–Schwarz gives

$$F_e = \frac{1}{M^2}\sum_\ell |\,\mathrm{tr}\,A_\ell|^2 \le \frac{1}{M}\sum_{j,\ell} |(A_\ell)_{jj}|^2 \le P_{\mathrm{c}}^*(V). \tag{S12}$$

The middle expression is the success probability for that recovery followed by a logical-basis measurement. Maximizing over recoveries proves $F_{\mathrm{q}}^*(V) \le P_{\mathrm{c}}^*(V)$ pointwise. Quantum achievability will close the gap between this ordering and the classical converse in the regular critical window.

AN EXPLICIT QUANTUM RECOVERY FOR LIKELY ERRORS

Let $\mathcal{G} = \{P : s(P) < n-k-a\}$ and $L = |\mathcal{G}|$. Its probability threshold implies $L < d2^{-a}/M$. If $\mathcal{G}$ is empty the claimed lower bound is trivial. Otherwise define

$$B_P = PV, \qquad S = \sum_{P\in\mathcal{G}} B_P B_P^\dagger, \qquad R_P = B_P^\dagger S^{-1/2}. \tag{S13}$$

All inverses are on the support of S. Then $\sum_P R_P^\dagger R_P = \Pi_{\mathrm{supp}S}$. If $\{|e_b\rangle\}$ is an orthonormal basis of $\ker S$, the additional Kraus operators $K_b = |0_{\mathrm{L}}\rangle\langle e_b|$ complete a trace-preserving decoder. The label P indexes a Kraus representation of this fixed decoder; it is not a supplied record of the actual noise event. The decoder is a transpose/square-root recovery for the uniformly weighted truncated error family [1].

Let $|v_{jP}\rangle = B_P|j\rangle$ and $a_{jP} = \langle v_{jP}|S^{-1/2}|v_{jP}\rangle$. The fine-grained square-root POVM has effects

$$E_{jP} = S^{-1/2}|v_{jP}\rangle\langle v_{jP}|S^{-1/2}, \qquad \sum_{j,P} E_{jP} = \Pi_{\mathrm{supp}S}. \tag{S14}$$

Consequently $0 \le a_{jP} \le 1$. Applying the Hayashi–Nagaoka inequality [2] with the rank-one projector onto v_{jP} as its first argument gives

$$1 - a_{jP}^2 \le 4C_{jP}, \qquad C_{jP} = \sum_{(i,Q)\neq(j,P)} |\langle v_{jP}|v_{iQ}\rangle|^2. \tag{S15}$$

The term $2(I - |v_{jP}\rangle\langle v_{jP}|)$ has zero expectation in v_{jP}.

The Kraus operators of the recovered noisy channel include $\sqrt{p_Q}R_PQV$. Retaining only terms with $Q = P \in \mathcal{G}$ in its nonnegative fidelity sum gives

$$F_e \ge \sum_{P\in\mathcal{G}} p_P \left(\frac{1}{M}\sum_j a_{jP}\right)^2 \tag{S16}$$

$$\ge p(\mathcal{G}) - 8X(U), \qquad X(U) = \sum_{P\in\mathcal{G}} \frac{p_P}{M}\sum_j C_{jP}. \tag{S17}$$

Here $(\overline{a})^2 \ge 1 - 2\overline{(1-a)} \ge 1 - 2\overline{(1-a^2)}$. The recovery is unweighted, but its performance is evaluated with the true error probabilities; these need not be equal.

Same-error columns are exactly orthogonal. Equation (S7) therefore implies

$$\mathbb{E}C_{jP} = \frac{L-1}{d+1} + \frac{(M-1)(L-1)d}{d^2-1} = \frac{(L-1)(dM-1)}{d^2-1} \le \frac{M(L-1)}{d}, \tag{S18}$$

where the inequality uses $M \le d$. Thus $\mathbb{E}X \le p(\mathcal{G})ML/d \le 2^{-a}$, and

$$\mathbb{E}F_{\mathrm{q}}^* \ge \mathcal{F}(n-k-a) - 8\,2^{-a}. \tag{S19}$$

A sharper bound for an arbitrary nonempty $\mathcal{G}$ follows from the same computation:

$$\mathbb{E}F_{\mathrm{q}}^* \ge p(\mathcal{G})\left[1 - \frac{8(|\mathcal{G}|-1)(dM-1)}{d^2-1}\right]. \tag{S20}$$

It can be more useful at small sizes, although a negative right-hand side is simply uninformative.

PROBABILITY OVER ENCODERS AND SECOND-ORDER CONSEQUENCES

Set $F_- = \mathcal{F}(n-k-a)$ and $F_+ = \mathcal{F}(n-k+2a)$. Equation (S17) implies

$$\mathbb{E}(F_- - F_{\mathrm{q}}^*)_+ \le 8\,2^{-a}. \tag{S21}$$

The pointwise proof of Eq. (S11), with $\gamma = M2^{-a}$, gives

$$\mathbb{E}(P_{\mathrm{c}}^* - F_+)_+ \le \frac{5}{4}\,2^{-a} + \frac{2^a}{4M}. \tag{S22}$$

Positive-deviation control places both tasks on a common probability event. Markov's inequality and a union bound give, for every $\eta > 0$,

$$\mathbb{P}_U[F_{\mathrm{q}}^* < F_- - \eta \text{ or } P_{\mathrm{c}}^* > F_+ + \eta] \le \frac{37\,2^{-a}}{4\eta} + \frac{2^a}{4M\eta}. \tag{S23}$$

Together with Eq. (S12), this brackets both operational quantities for the same realized encoder.

Now fix a single-qubit distribution $\boldsymbol{p}$ with $0 < H < 1$ and $\sigma^2 > 0$, where

$$H = -\sum_\alpha p_\alpha \log_2 p_\alpha, \qquad \sigma^2 = \sum_\alpha p_\alpha(-\log_2 p_\alpha - H)^2. \tag{S24}$$

Errors are i.i.d., so $s(P)$ is a sum of n independent variables. The third absolute centered moment is finite, even if some error probabilities vanish (zero-probability symbols are omitted). For

$$k_n = n(1-H) + \sqrt{n}\,\sigma z + O(1) \tag{S25}$$

with fixed z, M_n grows exponentially and remains below d eventually. Berry–Esseen [3], including its left-limit version for the strict CDF, gives

$$\mathcal{F}(n-k_n+b)=\Phi\left(-z+\frac{b}{\sqrt{n}\sigma}+O(n^{-1/2})\right)+O(n^{-1/2}). \tag{S26}$$

Choose $a=2\log_2 n$. The two CDF shifts are $O(\log n/\sqrt{n})$, and $2^a/M_n$ is exponentially small. This proves

$$\mathbb{E}F_{\mathrm{q}}^*=\Phi(-z)+O(\log n/\sqrt{n}),\qquad \mathbb{E}P_{\mathrm{c}}^*=\Phi(-z)+O(\log n/\sqrt{n}). \tag{S27}$$

Constants depend on the fixed distribution and on a bounded interval of z, not on which exact 2-design is chosen. With $\eta=n^{-1/2}$, the exceptional probability in Eq. (S23) is at most

$$\frac{37}{4}n^{-3/2}+\frac{n^{5/2}}{4M_n}. \tag{S28}$$

Thus both optimal performances have the same Gaussian limit in probability, and on the complement of this event differ by at most $O(\log n/\sqrt{n})$. This is a typical-encoder limit: finite n permits unequal performances, and the exceptional set is retained.

For fixed $0<\epsilon<1$ and sufficiently large n, define the largest logical sizes meeting the ensemble-mean error constraint by

$$k_{\epsilon,n}^{\mathrm{c}}=\max\{k\in\{1,\dots,n\}:\mathbb{E}_U P_{\mathrm{c}}^*(V_{n,k})\geq 1-\epsilon\}, \tag{S29}$$

$$k_{\epsilon,n}^{\mathrm{q}}=\max\{k\in\{1,\dots,n\}:\mathbb{E}_U F_{\mathrm{q}}^*(V_{n,k})\geq 1-\epsilon\}. \tag{S30}$$

The exact 2-design ensemble is prescribed for each (n,k), and the sets are nonempty for sufficiently large n under the stated assumptions. Thus the constraint is on the average of separately optimized decoder performances, rather than an optimization over encoder realizations. The finite bracket is uniform in k. Evaluating its lower and upper sides at $k=n(1-H)+\sqrt{n}\sigma\Phi^{-1}(\epsilon)\mp C\log n$ and using the positive Gaussian density at the fixed quantile proves

$$k_{\epsilon,n}^{\mathrm{c}}=n(1-H)+\sqrt{n}\sigma\Phi^{-1}(\epsilon)+O(\log n),\qquad k_{\epsilon,n}^{\mathrm{q}}=n(1-H)+\sqrt{n}\sigma\Phi^{-1}(\epsilon)+O(\log n). \tag{S31}$$

The spectrum upper bound decreases with k at fixed a, which bounds all larger sizes; no unproved monotonicity of an arbitrarily specified code family is needed.

For fixed positive rate $R=k/n$, taking $a=\alpha n$ with $\alpha>0$ sufficiently small yields exponential approach to one if $R<1-H$, and exponential decay if $R>\max(1-H,0)$. This follows from classical large deviations for the finite-alphabet surprisal. The finite bracket also covers $H=1$, fixed M, and zero surprisal variance, which lie outside Eq. (S27). The term $2^a/(4M)$ displays the growing-message-space requirement for the common Gaussian curve.

CLIFFORD DECODING: EXACT CLASSICAL FORMULAS FOR BOTH OPTIMA

For this section take the standard embedding $V_0|j\rangle=|j\rangle_{\mathrm{L}}|0\rangle_{\mathrm{A}}$ with $M=2^k$, and let U be Clifford. Conjugating a physical Pauli through the known inverse encoder produces a Pauli on logical and auxiliary qubits. Denote its logical X and Z labels by $x,z\in\{0,1\}^k$ and its auxiliary X label by $s\in\{0,1\}^{n-k}$. Auxiliary Z labels only contribute phases on $|0\rangle_{\mathrm{A}}$ and are summed. Let $w(s,x,z)$ be the resulting joint probability, including all Pauli degeneracies.

The auxiliary register is in the orthogonal computational state $|s\rangle$. Conditional on s, the logical channel is a Pauli channel with unnormalized probabilities $w(s,x,z)$. For a decoder on that syndrome block with Kraus operators D_ℓ, define

$$f_{x,z}=\frac{1}{M^2}\sum_\ell|\operatorname{tr}[D_\ell X^x Z^z]|^2\geq 0. \tag{S32}$$

Pauli orthogonality and trace preservation imply $\sum_{x,z}f_{x,z}=1$. Thus the conditional contribution to entanglement fidelity is at most $\max_{x,z}w(s,x,z)$, attained by correcting the most probable logical Pauli. Since syndrome blocks are orthogonal, an arbitrary decoder cannot improve by coherently mixing them. Therefore

$$F_{\mathrm{q}}^*=\sum_s\max_{x,z}w(s,x,z). \tag{S33}$$

This syndrome/logical-coset representation follows the stabilizer formalism [4]. Optimal recovery by the most likely logical class appears in Sec. III B of Ref. [5]; it also underlies stabilizer-code converse results [6].

For a classical basis message j, the observed logical bit string after inverse encoding is $y = j \oplus x$. Phase labels z have no effect. All output states commute, so dephasing any POVM in this common basis preserves its success. For uniform messages the optimal rule chooses the most probable x at each syndrome, giving

$$P_{\mathrm{c}}^{*} = \sum_{s} \max_{x} \sum_{z} w(s, x, z). \tag{S34}$$

A known inverse Clifford followed by computational-basis measurement and classical maximum-likelihood processing implements this optimum. Computing that processing need not be efficient. Encoder, inverse encoder, and syndrome measurements are ideal in this model.

NUMERICAL METHOD AND REPRODUCIBILITY

The implementation in `code/recovery.py` uses NumPy, SciPy, Qiskit, CVXPY, CLARABEL, and Matplotlib. Qiskit's `random_clifford` samples the Clifford group using the Bravyi–Maslov construction [7]. Each sampled inverse symplectic matrix and its seed are stored, preserving the realized ensemble independently of future sampler changes. The Pauli phase of a Clifford is immaterial to Eqs. (S33)–(S34).

We use $k/n = 1/4$ and $n = 4, 8, 12, 16$, with 64 independently sampled encoders at each size. Each encoder is reused throughout the noise scan. For depolarizing noise $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, the Fourier transform of a single physical Pauli probability is 1 at the trivial character and $1-r$ at each nontrivial character. The retained variable (s, x, z) has $n+k$ bits. Its binary dual map, obtained from the inverse Clifford symplectic matrix, maps each character to a physical Pauli support of weight w. The characteristic function is consequently $(1-r)^w$. An inverse Walsh–Hadamard transform computes the entire $w(s, x, z)$ distribution, summing the discarded auxiliary-Z labels exactly. Equations (S33) and (S34) then give both optima without a density-matrix approximation. Negative Fourier roundoff smaller than 2×10^{-12} is clipped and the distribution renormalized; larger negativity aborts the calculation.

Independent direct enumeration of all 4^n physical errors for $n \leq 3$ checks the conjugation convention and Fourier implementation. For $n = 2, 3$, we separately optimize classical POVMs and quantum recovery Choi matrices for Haar and Clifford encoders. Both primal and dual semidefinite programs are solved with CLARABEL. If it returns `optimal_inaccurate`, the same problem is independently solved with the installed SCS solver; all attempts and their statuses are retained. Only a final `optimal` status is accepted. Independently computed trace-preservation/completeness residuals, primal and dual minimum eigenvalues, and primal–dual objective gaps must satisfy an absolute 2×10^{-7} tolerance. The identity-channel endpoints are one; under complete depolarization the endpoints are $1/M$ for classical guessing and $1/M^2$ for entanglement fidelity. Explicit truncated square-root recoveries are checked for trace preservation and against both the analytical lower estimate and the optimal recovery SDP.

The large-n critical panel evaluates binomial CDFs in the proved bounds. The strict surprisal inequality is checked against neighboring error-weight levels; near a floating-point boundary we use 80-digit decimal comparison, with exact logarithms for binary-power probabilities. Regression checks include exact lattice thresholds and the immediately adjacent floating-point thresholds. The dataset and machine-readable verification report distinguish these analytic evaluations from sampled Clifford codes: the large-size curves are bound evaluations, not quantum decoding simulations. Sampling uncertainty is the standard error across encoders, $s/\sqrt{64}$; it describes the ensemble-mean estimate, not individual-code confidence or experimental noise. The calculations use ideal gates and do not include hardware experiments.

RELATION TO EARLIER RESULTS AND SCOPE OF PROOF CHECKING

The hashing rate and its use in random quantum coding are established results [8, 9]. A particularly direct precursor is Klesse's bound for a trace-decreasing completely positive operation [10]. In the notation of this manuscript, applying Eqs. (35) and (38) of its arXiv v2 to the truncated Pauli map $\mathcal{N}_{\mathcal{G}}$ gives

$$\mathbb{E}F_{\mathrm{q}}^{*}(\mathcal{N}_p) \geq \mathbb{E}F_{\mathrm{q}}^{*}(\mathcal{N}_{\mathcal{G}}) \geq q - \sqrt{ML}\,\|\mathcal{N}_{\mathcal{G}}(I/d)\|_2 = q\left(1 - \sqrt{ML/d}\right), \tag{S35}$$

where $q = p(\mathcal{G})$, $L = |\mathcal{G}|$, and $\|\cdot\|_2$ is the Hilbert–Schmidt norm. There is no normalization by the truncation probability. Thus the same surprisal cut gives $\mathbb{E}F_{\mathrm{q}}^{*} \geq \mathcal{F}(n-k-a) - 2^{-a/2}$, already sufficient for the Gaussian

achievable side at $a = 2\log_2 n$. Klesse states the result for Haar codes; its projection second moment, Eq. (26), also applies to exact unitary 2-designs, as discussed in Sec. VI.1 of Ref. [11]. Moreover, Klesse's pointwise Eq. (36) and its averaged estimate imply

$$\mathbb{E}\big(q - F_{\mathrm{q}}^*\big)_+ \le q\sqrt{ML/d} \le 2^{-a/2}. \tag{S36}$$

Markov's inequality then supplies quantum high-probability achievability. Taking $a = 4\log_2 n$ and $\eta = n^{-1/2}$ gives polynomially vanishing exceptional probability at the same $O(\log n/\sqrt{n})$ Gaussian accuracy. Our explicit recovery has truncation loss $8\,2^{-a}$. The quantum Gaussian limit, exact-design extension, and high-probability achievability thus have a direct route through Klesse's result; the classical optimal-decoding converse supplies the matching side needed to determine both tasks.

Renes [12] connects approximate quantum recovery to guessing two complementary observables. Its Eq. (3), in our overlap convention, reads $\arccos\sqrt{F_{\mathrm{q}}^*} \le \arccos P_Z + \arccos P_X$, where P_Z, P_X are the separately optimized guessing probabilities for two Fourier-conjugate logical bases. If both guessing errors are at most ϵ, it implies $1 - F_{\mathrm{q}}^* \le 8\epsilon + O(\epsilon^2)$ as $\epsilon \to 0$. This gives linear small-infidelity control; the square-root dependence below concerns a trace-distance bound. At fixed nonzero error the angle bound leaves a gap between the optimal classical and quantum performances. Nakata, Matsuura, and Koashi [13] give a complementary-basis decoder construction. Their Corollary 1, Eqs. (39)–(40) of arXiv v5, bounds its quantum trace-distance error by

$$\Delta_{\mathrm{q}} \le \sqrt{\Delta_{\mathrm{cl},E}(2 - \Delta_{\mathrm{cl},E})} + \sqrt{\Delta_{\mathrm{cl},F}} \le (1+\sqrt{2}) \max_{W=E,F} \sqrt{\Delta_{\mathrm{cl},W}}. \tag{S37}$$

Here E, F are mutually unbiased logical bases; the classical errors are average complete-message errors. Their quantum error is half the trace distance from the target maximally entangled state, whereas ours is the entanglement infidelity $1 - F_{\mathrm{q}}^*$. For any fixed recovery output ρ these metrics satisfy $1 - \langle\Phi_M|\rho|\Phi_M\rangle \le \frac{1}{2}\|\rho - \Phi_M\|_1 \le \sqrt{1 - \langle\Phi_M|\rho|\Phi_M\rangle}$. Proposition 4 of that work states a common classical and quantum recovery threshold in the Haar-random Hayden–Preskill model. Its following discussion also identifies random dynamics as a poor classical encoder compared with encodings tailored to classical messages. These are direct precedents for the common-threshold question and the encoding-cost interpretation.

There is a direct overlap with our noise model when the initial black hole is pure. In their notation, the encoder acts on $n = N + k$ qubits, of which ℓ remain accessible. Tracing out the other $m = n - \ell$ qubits is operationally equivalent to replacing them by a maximally mixed register: tracing that register out or appending it are receiver operations independent of the input. The replacement is the Pauli channel

$$\mathcal{T}_m(\rho) = \frac{1}{4^m} \sum_{P \in \{I,X,Y,Z\}^{\otimes m}} (P \otimes I)\rho(P^\dagger \otimes I) = \frac{I_{2^m}}{2^m} \otimes \mathrm{tr}_m\, \rho, \tag{S38}$$

after ordering the discarded qubits first. Its nonzero error probabilities are all 4^{-m}, so $s(P) = 2m$ with zero variance. Matching this spectral location to $n - k$ gives $\ell = (n+k)/2 = k + N/2$, the threshold in their pure-state example. This is a common spectral threshold, not an instance of our positive-variance i.i.d. Gaussian statement, and equality at the threshold is not implied by the finite bracket.

The classical optimal-success converse supplies the match left open by the complementary-basis bounds. For Pauli noise it meets quantum achievability and determines the leading Gaussian performance throughout the regular positive-rate window, the ensemble-mean second-order coding sizes, and a joint deviation bound for the same sampled encoder in any exact unitary 2-design. The probability event is defined for each fixed choice of parameters; a single encoder is not required to satisfy all choices simultaneously.

Design moments and probability control have established roles in random coding. Hayden *et al.* [14], Sec. 4, explicitly replace Haar random coding by any exact unitary 2-design. Dupuis *et al.* [15], Theorem 3.1 and Eq. (24), give a one-shot decoupling bound and its Markov high-probability consequence. Szehr *et al.* [16] treat approximate designs, with dimension-dependent error factors. The joint bracket here uses exact moments in the classical converse as well as quantum achievability; its extension to approximate designs remains open.

Information-spectrum converses [2, 17], classical second-order channel coding [18, 19], and quantum hypothesis testing [20, 21] resolve fluctuation-scale performance. Hayashi's Theorem 11 specifically expresses additive-channel second-order behavior through the noise self-information. The quantum hypothesis-testing results concern binary state discrimination, rather than the two recovery tasks here. Finite-resource quantum coding provides second-order achievability bounds and matching converses for some channels [22]. The published Appendix D of Ref. [23] develops smooth-entropy second-order achievability estimates for random Clifford block encodings; its arXiv version does not

contain that same appendix. Gullans *et al.* [5] analyze optimal quantum decoding of random stabilizer codes and low-depth circuit encodings. Their Sec. III B gives the logical-coset maximum-likelihood rule, and Appendix C derives an $n^{-1/2}$ critical-window width from fluctuations in the number of i.i.d. erasures within their random-matrix analysis. This establishes a direct precedent for quantum recovery transitions and fluctuation broadening, with erasure locations supplied to the decoder. The present Pauli model supplies no such flag and pairs optimal quantum recovery with classical complete-message decoding. Ref. [11] studies coherent information and spectra of Haar-random codes, including hashing transitions and a square-root finite-size window. Ref. [24] proves high-probability approximate correction by Haar codes for a bounded set of errors. Ref. [6] supplies a strong converse for full stabilizer-eigenspace codes under an input-dependent coherent-information condition.

The common Gaussian profile requires a fixed i.i.d. single-qubit distribution with $0 < H < 1$ and positive self-information variance. Its coding sizes refer to prescribed random-encoder ensembles, not the optimized quantum capacity of a general Pauli channel. Pairwise random-state contraction bounds [25] concern a different operational quantity. Ref. [26] studies measurement channels, mutual information, and accessible information in Sec. IV.3 and Appendix C, using the entire state-2-design alphabet in its state-ensemble statements. Our conditional alphabet, for each known U, is the set of M orthogonal columns $\{UV_0|j\rangle\}$ and is not itself a state 2-design. Optimizing the POVM for that alphabet gives $\mathbb{E}_U \max_{\{E_j^{(U)}\}} P_{\mathrm{succ}}(U)$. Averaging over U preserves this conditional optimization; it does not produce the accessible-information problem for a full design alphabet. Both the alphabet and the optimization order distinguish the tasks.

The recovery theorem, joint probability bound, and Gaussian coding result rest on the explicit proofs above, numerical consistency checks, and independent model-assisted mathematical review. The project's earlier Lean formalization covers conditional two-state information-spectrum inequalities and algebraic examples, not these recovery results.

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